--Part A: Multiple choice

-Be sure that you have 12 different questions. (24 - marks)
-Complete the table by putting (the correct answer letter) in the table

Q1) Three pen are drawn at random from 10 pen, of which 6 are secondband. What is the probability that get one secondband?

a) 0.6 b) 0.7 c) 0.3 d) 0.8

Q2) Two independent machines A, B., Working such that p(A)=0.2, p(B)=0.6. What is the probability that machine A works and machine B does not work

a) 0.06 b) 0.07 c) 0.08 d) 0.02

Q3) If A and B be two events in a random experiment such that $P(A)=0.3$, $P(B)=0.1$ and $P(A\cap B)=0.2$ find $P(\overline{A}\,|\,\overline{B})$

a) 7/9 b) 4/5 c)7/8 d) 8/9

Q4) Let A: diabetes, B: cholesterol, in a random experiment we find p(A)=0.5, p(B)=0.4 and p(A∩B)=0.3, a person is chosen at random, find the probability that he has diabetes such that he has cholesterol

a) 0.06 b) 0.95 c) 0.54 d) 0.75

Q5) Let X random variable on $[0,3]$ and f(x) =1/3 the Var(x) is

a) 3/7 b) 3/4 c) 4/5 d) 5/8

Q6) Let (X,Y) be a bivariate random variable such that $E(X)=2.5$, $\sigma_x=3$, $E(Y)=4$, $\sigma_y=2$ and $E(XY)=12$, then Corr (2x-1, 5-3y)

a)3/7 b) -1/3 c)1/6 d) 2/5

Q7) A student answers a multiple-choice examination of five problems by pure guessing. If there are 5 choices for each problem, what is the probability that he will get three correct answers?

a) 0.0510 b) 0.2611 c) 0.7406 d) 0.0181

Q8) If X is the number of cars passing through some road per day, if the average number of cars per day is 6, what is the probability that on a given day there will be exactly 5 cars passing?

a) 0.161 b) 0.038 c) 0.074 d) 0.838

Q9) The following table provides the joint probability distribution of X and Y

x \ y	1	2	3
1	0.10	0.30	0.1
2	0.1	0.2	0.2

The $P(x>1\,|\,y=3)=$ a) 2/3 b) 2/5 c) 2/7 d) 2/9

Q10) The prices of Vegetables this day are normally distributed with a mean 38 piaster and a standard deviation 4 piaster. If a day is taken at random, what is the probability that its price exceeds 43 piasters?

a) 0.156 b) 0.1056 c) 0.0156 d) 0.8401

Q11) If X-N(20, 25), then P(X>30) =

a) 0.00567 b) 0.3458 c) 0. 0228 d)0. 762

Q12) Given the auto-correlation function for a second-order stationary periodic process $R_{xx}(T)=36+\frac{3}{1+47^2}$ the variance of the process is.

a)-1 b)6 c) 4

Part B: Fill the Blanks Final answer (18 - marks)

Q13: Consider the distribution, with the probability density function $f_x(x)=e^{-x}$, $x>0$. Find the probability density function of y such that $Y=e^x-1$, $y>0$. $f_Y(Y) = (Y+1)^{-\theta-1}$

Q 14: If the probability that of heavy smokers will have lung cancer after the age of forty is 0.8, in a sample of heavy smokers with a mean 4, what is the probability that at most two heavy smokers will suffer from lung cancer? $P = 0.05792$

Q 15: If X is binomial random variable with mean 1.6 and variance 1.28 the $P(x=3) = 0.28$

Q 16: Let $X-poisson$ (2.5) the $P(1<x<4) = 0.470278$

Q 17: Let $X-poisson$ (2) the $E(2x^2-x+1) = 11$

Q 18: $X-N(\mu,\sigma^2)$ then $P(\mu+\sigma<X<\mu+2.05\sigma) = 0.13248$

(8 - marks)

Part C: Written Questions

Q 19: Show that the process $X(t)=\cos(t+\theta)$ is first-order stationary if θ is a uniformly distributed random variable on $(0, 2\pi)$. $E(x(t)) = \bar{x} = 0$

Q 20: The temperature grades of summer weather this year are normally distributed with a mean 40 degrees and a standard deviation of 2 degrees. If the greatest 33% of grades will be dangerous, what is the smallest grade that will be dangerous?

$X = 40.88 \approx 41$ degrees.

All the best -

PART (A):

Q1: 10 pens → 3 pens drawn $\quad \#\Omega = \binom{10}{3} = 120$

10 pens → 6 second band (s.b)
→ 4 not second band.

$$P(1 \text{ s.b } \& \text{ 2 not s.c}) = \frac{\binom{6}{1}\binom{4}{2}}{120} = \frac{36}{120} = 0.3$$

Answer : C.

Q2: $P(A) = 0.2$, $P(B) = 0.6$

$$P(A \cap \bar{B}) = P(A - B) = P(A) - P(A \cap B)$$

$$= 0.2 - 0.2 \times 0.6 = 0.2(1 - 0.6) = 0.08$$

Answer : C.

Q3: $P(A) = 0.3$, $P(B) = 0.1$, $P(A \cap B) = 0.2$

$$P(\bar{A} \mid \bar{B}) = \frac{P(\bar{A} \cap \bar{B})}{P(\bar{B})} = \frac{\overline{P(A \cup B)}}{1 - 0.1}$$

$$= \frac{1 - P(A \cap B)}{0.9} = \frac{1 - (P(A) + P(B) - P(A \cap B))}{0.9}$$

$$= \frac{1-(0.4-0.2)}{0.9} = \frac{1-0.2}{0.9} = \frac{0.8}{0.9} = \frac{8}{9}$$

Answer: D.

Q4: $P(A)=0.5$, $P(B)=0.4$, $P(A \cap B)=0.3$

such that ⟶ condition

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.4} = \frac{3}{4}$$

Answer: B.

Q5: X continuous r.v

$$f(x) = \begin{cases} 1/3, & 0 \le x \le 3 \\ 0, & \text{o.w} \end{cases}$$

$$\text{Var}(X) = EX^2 - (EX)^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}$$

Answer: B.

$$EX = \int_0^3 \frac{1}{3} x \, dx = \frac{x^2}{6}\Big]_0^3 = \frac{9}{6} = \frac{3}{2}$$

$$EX^2 = \int_0^3 \frac{x^2}{3} dx = \frac{x^3}{9}\Big]_0^3 = \frac{27}{9} = 3$$

Q6: $EX = 2.5$, $\text{std}\,X = \sigma_X = 3$
$EY = 4$, $\text{std}\,Y = \sigma_Y = 2$
$\rightarrow EXY = 12$

$$\text{Corr}(2X-1, 5-3Y) = -\text{Corr}(X,Y) = -\frac{\text{Var}(X,Y)}{\sigma_X \sigma_Y}$$

$$\Rightarrow \text{Var}(X,Y) = E(XY) - (EX)(EY) = 12 - 10 = 2$$

$$= -\frac{2}{3 \times 2} = -1/3$$

Answer: B.

Multiple choice EXAM:-

Q7: Q1: A. B. C. D. E.

⋮

Q5: ...

each question is an experiment

$p(\text{right answer}) = \frac{1}{5} = 0.2$

with fixed success & fail probabilities.

the experiment is repeated 5 times.

Therefore : $X \sim B(5, 0.2)$

$P(3 \text{ correct}) = P(X = 3)$

$= \binom{5}{3} \ 0.2^3 \ 0.8^2 = 0.0512$

Answer : A.

Q8 : X car passing per day

$X \sim \text{Poisson}(m)$

$\text{avg} = m = 6 : \ X \sim P(6)$

$P(X=5) = \dfrac{e^{-6}(6)^5}{5!} = 0.1606 \approx 0.161$

Answer : A.

Q9 : $P(X > 1 \mid Y = 3) = P(X = 2 \mid Y = 3)$

$= \dfrac{P(X=2 \cap Y=3)}{P(Y=3)}$

- $p(X=2 \cap Y=3) = 0.2$

X \ Y	1	2	3
1	0.10	0.30	0.1
2	0.1	0.2	0.2

- $p(X=3) = 0.1 + 0.2 = 0.3$

X \ Y	1	2	3
1	0.10	0.30	0.1
2	0.1	0.2	0.2

So, $p(X>1 \mid Y=3) = \frac{0.2}{0.3} = 2/3$

Answer: A.

Q10: X : prices of vegatiables. ($ piasters يعني قروش)

mean $= \mu = 38$, std $X = \sigma x = 4$.

exceeds ⟶ تزيد عن

$p(X>43) = 1 - p(X \leq 43)$, $z = \frac{43-38}{4} = 1.25$

$= 1 - p(z \leq 1.25) = 1 - 0.89435 = 0.10565$

~ using z- table ~ Answer: B.

Q11: $X \sim N(20, 25) \longrightarrow \mu = 20, \sigma = +5$

$P(X > 30) = 1 - P(X < 30) \longrightarrow z = \frac{30-20}{5} = 2$

$= 1 - P(z < 2) = 1 - 0.97725 = 0.02275$

~ By z-table ~ ≈ 0.0228

Answer: C.

Q12: $C_{XX}(T) = R_{XX}(T=0) - \lim_{T \to \infty} R_{XX}(T)$

$= 36 + 3 - 36 = 3$

Answer: D.

PART (B):

Q13: $f_X(x) = e^{-\theta x}, \quad x > 0$

$Y = e^x - 1 \longrightarrow x = \ln(Y+1)$

$\frac{dx}{dy} = \frac{1}{Y+1}$

by transformation:

$f_Y(Y) = f_X(T^{-1}(y)) \left| \frac{dx}{dy} \right|$

$$f_Y(y) = f_X(\ln(y+1)) \left| \frac{1}{y+1} \right| = \frac{(y+1)^{-\theta}}{|y+1|}$$

for $(y > 0) \Rightarrow f_Y(y) = (y+1)^{-\theta-1}$ #

Q14: heavy smokers having lung cancer after 40: $p = 0.8$, $q = 0.2$.

$$EX = 4 = np \longrightarrow n = \frac{4}{0.8} = 5$$

$$X \sim B(5, 0.8)$$

$$p(\text{2 at most}) = p(X=2) + p(X=1) + p(X=0)$$

$$= \binom{5}{2} 0.8^2 0.2^3 + \binom{5}{1} 0.8\, 0.2^4 + \binom{5}{0} 0.8^0 0.2^5$$

$$= 0.2^3 \left(10\ 0.8^2 + 5\ 0.8\ 0.2 + 0.2^2 \right)$$

$$= 0.008\ (6.4 + 0.8 + 0.04) = 0.008 \times 7.24$$

$$= 0.05792 \quad \#$$

Q15:

$$\frac{Var(X)}{EX} = \frac{qnp}{np} = q = \frac{1.28}{1.6} = \frac{4}{5} = 0.8$$

$$p = 1 - 0.8 = 0.2 \longrightarrow EX = np \longrightarrow n = \frac{EX}{p} = \frac{1.6}{0.2} = 8$$

$$X \sim B(8, 0.2)$$

$$\rightsquigarrow p(X=8) = \binom{8}{8} 0.2^8 \, 0.8^0 = 0.2^8 = 2.56 \times 10^{-6}$$

Q16:

$$p(1<X<4) = p(X=2) + p(X=3)$$

$$= \frac{e^{-2.5}(2.5)^2}{2!} + \frac{e^{-2.5}(2.5)^3}{3!}$$

$$= \frac{(2.5)^2 e^{-2.5}}{2!}\left(1 + \frac{2.5}{3}\right) = \frac{11(2.5)^2 e^{-2.5}}{12}$$

$$= 0.470278 \approx 0.47028 \ \#$$

Q17:

$$E(2X^2 - X + 1) = 2EX^2 - EX + 1$$

- $EX^2 = m^2 + m = 4 + 2 = 6$, $EX = m = 2$

$$= 2(6) - 2 + 1 = 10 + 1 = 11 \ \#$$

Q18: $P(\mu+\sigma < X < \mu+2.05\sigma)$

$z_1 = \frac{\mu+\sigma-\mu}{\sigma} = 1$ ↵ ↳ $z_2 = \frac{\mu+2.05\sigma-\mu}{\sigma} = 2.05$

$P(1 < Z < 2.05) = P(Z < 2.05) - P(Z < 1)$

~using z-table~ $= 0.97382 - 0.84134$

$= 0.13248$ #

PART C:

Q19: $\theta \sim U(0, 2\pi)$, $f(\theta) = \begin{cases} \frac{1}{2\pi}, & 0 \le \theta \le 2\pi \\ 0, & o.w \end{cases}$

$X(t) = \cos(t+\theta)$

- 1st order-stationary: $E(X(t)) = \overline{X} \equiv$ ثابت

$$E(X(t)) = \int_{-\infty}^{\infty} X(t)\, f(\theta)\, d\theta = \int_{0}^{2\pi} \frac{A}{2\pi} \cos(t+\theta)\, d\theta$$

$$= \frac{A}{2\pi} \sin(t+\theta)\Big|_0^{2\pi} = \frac{A}{2\pi}\left(\sin(t+2\pi) - \sin t\right)$$

- $\sin(a+b) = \sin a \cos b + \cos a \sin b$

$\sin(t+2\pi) = \sin t \cos 2\pi + \cos t \sin 2\pi$

$\cos 2\pi = 1$, $\sin 2\pi = 0$

$\sin(t+2\pi) = -\sin t$

$$\Rightarrow \frac{A}{2\pi}(-\sin t + \sin t) = 0 = \bar{X} \quad \#$$

Q20: $\mu = 40$, $\text{std}X = \sigma_x = 2$

p greatest : 0.33

p smallest : $p(X < x_1) = 1 - 0.33 = 0.67$

$p(z < z_1) = 0.67 \longrightarrow z_1 = 0.44$

~ using z-table ~

$x_1 = \sigma z_1 + \mu = 2(0.44) + 40 = 40.88 \approx 41$ deg.

:: The End ::